Western Sydney

CORRELATION ON STOCK

THE JOURNAL

2023

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Economic and Financial Modeling

Effect of Macro-economic Indicators and Financial Information on companies' SP in Vietnam stock market: Panel Data Analysis from 2012-2022

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EFM-T323WSB-4

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1st December 2023

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C. Abbreviation

CAPEX: Capital Expenditures.

CF: Cash Flow.

CPI: Consumer Price Index.

DPR: Dividend Payout Ratio.

EG: Economic Growth.

EPS: Earnings Per Share.

FS: Financial Statements.

GDP: Gross Domestic Product.

GLS: Generalized Least Squares.

INF: Inflation.

M1: Money Supply M1 (cash and checking deposits).

M&A: Mergers and Acquisitions.

MLR: Multiple Linear Regression.

NI: Net Income.

NR: Net Revenue.

OCF: Operating Cash Flow.

OPEX: Operating Expenses.

P/B: Price to Book ratio.

P/E: Price to Earnings ratio.

REM: Random Effect Model.

RIR: Real Interest Rate.

SE: Standard Error.

SP: Stock Price.

TA: Total Assets.

TFA: Total Fixed Assets.

D. Abstract

The research aims to investigate the impact of macroeconomics variables, financial data and technical factors on the SP of companies in the Vietnam market by randomly sampling on 19 companies on the Vietnam VN30-Index from 2012 to 2022. A panel regression model is used which examined both financial (banks and insurance institutions) and non-financial companies in the time length of 11 years in the Vietnam Stock Market. The data source of this research is primary and secondary data, which are macro-economic data and financial information for the period from the World Bank, companies' financial statements and Bank's Report. The results provide strong evidence that INF, GDP, NR and Dividend Paid Activity is significantly correlated with companies' SPs in the Vietnam Market. Although this paper presents divergent findings compared to other studies, it is significant that the observed correlations between macro variables, financial information, and dummy variables remain relevant to SP.

E. Introduction

In Vietnam, the stock market, a crucial economic component, has been operational for over two decades. It wields significant influence on corporate and industrial expansion, potentially affecting the broader health of the economy by providing a platform for businesses to raise capital, fostering growth, and bolstering employment, which in turn enhances productivity. However, the relationship between stock market performance and EG has been debated, revealing three potential trends: a positive link, a negative link, or a bidirectional causative relationship. Some studies indicate a lack of correlation between these two factors. Duda (2020) contends that stock market indices may not inherently drive EG, echoing Lucas's (1988) view that the stock market is inconsequential to financial systems and economic development, suggesting minimal support for EG from the stock market.

Since its establishment in 2000, Vietnam's capital market has expanded to meet the increasing demands of investors and stakeholders. The Vietnam stock exchange has witnessed fluctuations but has remained integral to the economy by providing medium to long-term capital for the government and enterprises, thus being a crucial channel for economic capital mobilization. However, the literature on the stock market's influence on EG remains scant. A notable study by

Nguyen (2019) posits a two-way relationship between stock market development and EG in Vietnam, observable in both the long and short term.

This research aims to explore the connection between macroeconomic indicators and corporate financial data on SPs in Vietnam's market from 2012 to 2022, striving to establish the importance of stock market development in influencing EG.

The paper progresses by reviewing relevant literature in Section B and detailing the model specifications affecting SPs in Section C. Section D outlines the data and methodologies used in the study, while Section E delves into the empirical findings. Section F concludes the paper.

F. Literature Review

I. RIR and INF

According to Mulyanti et al. (2019), both RIR and INF have negative and significant effects on SP. The research explains that greater RIR would require companies to spend more on interest payments, reducing enterprises' net profit. Consequently, the companies' stock is less attractive to investors. Additionally, when INF rises uncontrollably, companies face higher operating costs, leading to a reduction in net profit and ultimately a decline in SP. The idea is supported by the findings of Al-Shubiri (2010).

In addition, an increase in consumer prices will cause INF (Mankiw, 2018). In other words, CPI and INF have a positive relationship in an economy. Therefore, based on Nasseh and Strauss (2000) research, CPI has a positive correlation with SP, which means that INF is positively correlated with SP, showcasing the opposite result in research articles on this issue.

II. GDP

According to research conducted by Singh et al (2011), it was observed that there is a positive correlation between GDP and SPs. This means that as the GDP increases, indicators such as P/E, P/B, and yield of return also tend to increase. The overall finding of the study suggests that the

growth of GDP is likely to result in higher SP for medium and large companies. However, it should be noted that there is a negative correlation between GDP and SP for small companies in Taiwan.

In various studies, it has been found that there is a strong and lasting relationship between SP and GDP. Additionally, there is a dynamic and positive connection between GDP and SP. This implies that when SP increase, the GDP of Turkey will also experience a rise and vice versa. This finding has been supported by research conducted by Kim and In (2003), Cole et al. (2008), Beck and Levine (2004), and Zhou et al. (2012).

III. M1

Previous research has demonstrated the influence of M1 on SP of companies which has shown that there exists an inverse relationship between M1 and RIR, as evidenced by Stracca (2003). This negative correlation between RIR and SP implies a positive relationship between M1 and SP.

IV. NR

Due to research by Srinivasan (2012), the size of the firm - which is measured by total NR- stated that it has a significant impact on companies' SP of all sectors, not including manufacturing. In detail, a company's NV is positively correlated with its SP. That is, the growth of NR will promote the increase in SP.

V. CAPEX

Jiang et al. (2006) figured out that there is a positive relation between CAPEX and corporation earnings of listed companies in Shanghai. This means that CAPEX keeps an important position in developing business operations, and value of SP. Furthermore, in the same study, enterprise earnings measured as the ratio of OCF to TA increased portfolio performance. Therefore, this paper has put this operating factor for considering correlated variables to SP.

VI. ROAA, ROAE and EPS

The impact of ROAA and ROAE on SP is determined to be significant based on the findings of the MLR Analysis and Hypothesis testing using t-test, f-test, and coefficient of determination (Bayhaqiyi, 2022). A different study conducted by Mulyanti et al. (2019) suggests that ROAA has a significant and positive influence on SP by enhancing a company's asset efficiency.

In a study conducted by Fitriana R. (2022), which analyzed non-financial companies over a period of 5 years, it was found that EPS explains the SP, while the DPR has either a detrimental effect or no effect on the SP. This was tested using the BPR model, which is the same model used in this research. Another study conducted by Bayhaqiyi et al. (2022), covering a period of 8 years, found that EPS does not have a significant effect on SP based on partial (t-test) analysis, but it does influence SP when analyzed simultaneously (f-test).

G. Specification of the Model

I. Research Objectives

The model is used to test the effects of the elements below which includes both macroeconomic indicators and internal company information on the SP of companies: RIR, GDP, INF, M1, NR, CAPEX, ROAA, ROAE, the activity of paying dividend to shareholders by money, The activity of M&A (Both buying and selling) and EPS.

II. Model Hypothesis

1. **RIR**

$H_0: \beta_1 = 0$ – No effect on SP of RIR

$H_1: \beta_1 \neq 0$ - Having effect on SP of RIR

2. **GDP**

$H_0: \beta_2 = 0$ - No effect on SP of GDP

$H_1: \beta_2 \neq 0$ - Having effect on SP of GDP

3. **INF**

$H_0: \beta_3 = 0$ - No effect on SP of INF

$H_1: \beta_3 \neq 0$ - Having effect on SP of INF

4. **M1**

$H_0: \beta_4 = 0$ - No effect on SP of M1

$H_1: \beta_4 \neq 0$ - Having effect on SP of M1

5. **NR**

$H_0: \beta_5 = 0$ - No effect on SP of NR

$H_1: \beta_5 \neq 0$ - Having effect on SP of NR

6. **CAPEX**

$H_0: \beta_6 = 0$ - No effect on SP of CAPEX

$H_1: \beta_6 \neq 0$ - Having effect on SP of CAPEX

7. **ROAA**

$H_0: \beta_7 = 0$ - No effect on SP of ROAA

$H_1: \beta_7 \neq 0$ - Having effect on SP of ROAA

8. **ROAE**

$H_0: \beta_8 = 0$ - No effect on SP of ROAE

$H_1: \beta_8 \neq 0$ - Having effect on SP of ROAE

9. **Paying Dividend Activity by Money**

$H_0: \beta_9 = 0$ - No effect on SP of Paying Dividend Activity by Money

$H_1: \beta_9 \neq 0$ - Having effect on SP of Paying Dividend Activity by Money

10. M&A Activity

$H_0: \beta_{10} = 0$ - No effect on SP of M&A Activity

$H_1: \beta_{10} \neq 0$ - No effect on SP of M&A Activity

11. EPS

$H_0: \beta_{11} = 0$ - No effect on SP of EPS

$H_1: \beta_{11} \neq 0$ - Having effect on SP of EPS

III. The Model

$$log(p_{i,j}) = \alpha + \beta_1 \times r_i + \beta_2 \times log(gdp_i) + \beta_3 \times infla_i + \beta_4 \times log(m1_i) + \beta_5 \times log(rev_{i,j}) + \beta_6 \times log(capex_{i,j}) + \beta_7 \times roaa_{i,j} + \beta_8 \times roae_{i,j} + \beta_9 \times divbm_{i,j} + \beta_{10} \times ma_{i,j} + \beta_{11} \times log(eps_{i,j}) + \mu_{i,j}$$

Where:

- **General symbol**

α: Overall intercept

$\beta_{1\text{-}11}$: Beta coefficience

log(x): Logarithmic function

i: Statistic year

j: Companies' code in HOSE

- **Dependent variable**

p: SP of companies

- **Independent variables**

r: RIR stated by the State bank of Vietnam.

gdp: GDP of Vietnam

inf: INF of Vietnam

m1: M1 of Vietnam

rev: NR of companies

capex: CAPEX of companies

roaa: ROAA of companies.

roae: ROAE of companies.

eps: EPS of companies.

- **Dummy variables**

divbm: Paying dividend by money = 1; Not pay by money = 0

ma: Having M&A activities (both selling and buying) = 1; Not having = 0

IV. Selection reason of independent variables

1. RIR

Previous studies were conducted in Pacific Island countries and South Africa. A specific research article for the Vietnamese market to analyze the impact of RIR on changes in SP is necessary. Furthermore, the model is developed to analyze the direct correlation between SP and RIR.

2. **GDP**

The above study is used to analyze the impact of GDP on investments. Therefore, to prove whether the change in SP is based on the growth of GDP or not, this model is performed.

3. **INF**

This study has included INF to examine the correlation of it with SP, more specifically the Vietnamese stock market.

4. **M1**

To show more correlation, this study analyzes the impact of an increase in M1 on the SP of companies in the same period instead of one subject. Meanwhile, the log-log function will reduce the large difference in units between the SP and M1.

5. **NR**

The study was developed to analyze how NR's growth is correlated with SP in the Vietnam market.

6. **CAPEX:**

To check the correlation between CAPEX and SP, the model is implemented. Specifically, effective investment in TFA can bring NR's growth which promotes the growth of SP.

7. **ROAA, ROAE, and EPS**

To clearly analyze its impact in diverse fields, ROA, ROE and EPS are added to the model. Both ROA&ROE assessing how well a company's management team is handling the funds entrusted to it, which directly affect companies' stock price. In addition, EPS which calculated based on net income and share outstanding, is included to confirm how it correlates to the SP.

VII. Expected Signs of Independent Variables on SP

Variable's Name	Explanation (Appendix 2)	Expected Sign
r	RIR ↑	Negative

	Investment ↓ Demand on stock ↓ SP↓	
gdp	GDP ↑ Investment ↑ Demand on stock ↑ SP↑	Positive
infla	Idea 1: INF ↑ SP ↑ Idea 2: INF ↓ SP ↓	Positive and negative
m1	RIR ↑ M1 ↓ Investment ↓ Demand on stock ↓ SP↓	Positive
rev	NR ↑ Profit may ↑ DPR ↑ Demand on stock ↑ SP ↑	Positive
capex	CAPEX ↑ NV ↑	Positive

	SP ↑	
roaa	NR ↑ Profit may ↑ ROAA ↑ SP ↑	Positive
roae	NR ↑ Profit may ↑ ROAE ↑ SP ↑	Positive
eps	Profit ↑ EPS ↑ SP ↑	Positive
divbm (Dummy variable)	Paying dividend Demand on stock ↑ SP ↑	Positive
ma (Dummy variable)	M&A announcement ↑ SP ↑	Positive

Table 1: Expected Signs of Independent Variables on SP

H. Data Description

Variables	Calculations & Definition	Sources
SP	Market SPs in Vietnam from 2012 to 2022	Vietstock

INF	An RIR that has been modified to take inflation into account.	World Bank
M1	M1 consists of demand deposits and checking accounts, which are the most widely utilized exchange mediums via debit cards and ATMs.	Asian Development Bank
GDP	Composed of goods and services produced for market sale as well as some non-market outputs provided by the government, such as defense or education services.	World Bank
CAPEX	The expense of investing in, updating, and maintaining fixed assets such as factories, machinery, and equipment... It reflects the CF of organizations that invest in acquiring fixed assets.	FS
NR	Net NR is typically recorded when a commission must be recognized, when a supplier receives a portion of the sales income, or when one party delivers clients for another party.	FS
ROAA	It is a measure of how well a corporation uses its assets.	Vietstock
ROAE	It is a metric of profitability. Since the mismatch between the income statement and the balance sheet, the metric is frequently calculated using average equity over a period.	Vietstock
M&A	Financial transactions between corporations that result in the consolidation of companies or their primary commercial assets.	FS
EPS	EPS indicates a company's net income allocated to each share of ordinary stock.	Vietstock

Dividend Paid Behaviour	The listed companies in the Vietnam stock market pay dividends or not.	FS

Table 2: Data description

I. Empirical Results

I. Hausman test (Detail on Appendix 03)

H_0: Difference in coefficient not systematic

	Result
Chi2 (10)	15.12
Prob>Chi2	0.1276
Conclusion	**Fail to reject H_0**

Table 3: The brief result of Hausman Test

The chi-squared statistic for the test is chi2(10) = 15.12 with a p-value of 0.1276. Since the p-value is greater than common significance levels (0.05, 0.01), we do not reject H_0 suggesting that **the REM may be an appropriate choice for the data**. However, the rank of the differenced variance matrix (10) does not equal the number of coefficients being tested (11), indicating a potential issue with the test. This means that the assumptions necessary for the Hausman test are not fully satisfied.

II. Basic Statistic

	R-square
Within	0.6130
Between	0.1308
Overall	**0.2847**

Table 4: R-squares of random REM

lnp	SE	z
r	0.0092263	-0.68
infla	0.0082329	-0.87
lngdp	0.2047389	5.13
lnm1	0.0742066	1.57
lnrev	0.0673083	4.19
lncapex	0.0311267	2.38
eps	0.0000161	0.55
divbm	0.0500628	1.39
ma	0.0332716	-2.10
roae	0.0035014	0.87
Roaa	0.0050625	-0.15
cons	1.585532	-6.66

Table 5: Standard error and z-value of REM

III. Testing

1. Heteroscedasticity Wald Test

H_o: No heteroscedasticity in REM.

Wald test	**Result**
LogE2 = X	17.5123
P-value	0.0000
Conclusion	**Reject H_0**

Table 6: Brief Heteroscedasticity Wald Test's Result for REM

Given that the p-value is 0.0000, it is less than any conventional significance level (such as 0.01 and 0.05). Therefore, we reject the null hypothesis of no heteroscedasticity. This leads to the conclusion that there is significant heteroscedasticity within the REM Thus, we have used robust

standard errors to correct for heteroscedasticity ensures more reliable statistical inference in the next step.

2. **Breusch and Pagan LM test (Detail on Appendix 04)**

H_0: Adopting the REM

Breusch and Pagan LM Test	Result
chibar2(01)	439.72
Prob > chibar2	0.0000
Conclusion	**Reject H_0**

Table 7: Brief Breusch and Pagan LM Test Result for REM

From the results of the model, the chi-squared value is 439.72 and the p-value is 0.0000. This means significant variations exist among the VN30 companies that the included variables do not fully explain. As a result, our analysis adopts a REM to incorporate the unobserved diversity among these entities. This approach enhances precision in assessing how the specified economic and financial factors influence VN30 stock market trends and Vietnam's EG.

3. **Autocorrelation Tests (Detail on Appendix 04)**

H_0: No serial correlation in the residuals of regression model

Serial Correlation Test	Result
ALM (lambda=0)	15.0800
p-value	0.0000
Conclusion	**Reject H_0**

Table 8: Autocorrelation Test Result for REM

With ALM value = 15.08 and p-value = 0, the test must reject the H_0 which means that there is no existence of autocorrelation. In other words, autocorrelation occurs between observations.

Their individual performance may be influenced by factors not captured in the model, and these persist over time (as indicated by the serial correlation). Therefore, the presence of serial

correlation must therefore be accounted for in the model to ensure accurate and reliable results. This might involve using GLS regression for panel data with a heteroskedasticity and autocorrelation (HAC) structure.

IV. Correction for Violation

1. Robust's SE

After detecting two assumption violations, heteroskedasticity and autocorrelation, the REM with Robust heteroskedasticity is applied. Consequently, for statistically significant variables (p-value < α = 0.05), SE increases after the Robust model is applied. P-value changes but it does not change the statistical significance of the independent variables.

	SE			
	Random Effect	Random Effect with heteroskedasticity robust	Differences	New P-value
r	0.0092263	0.0068228	-0.0024035	
infla	0.0082329	0.0058117	-0.0024212	
lngdp	0.2047389	0.03222985	0.1175596	0.001
lnm1	0.0742066	00783934	0.0041868	
lnrev	0.0673083	0.1005206	0.332123	0.005
lncapex	0.0311267	0.0365698	0.0054431	0.043
eps	0.0000161	0.0000242	0.000081	
divbm	0.0500628	0.0918917	0.0418289	
ma	0.0332716	0.040019	0.0067474	
ROAAe	0.0035014	0.0043768	0.0008754	
ROAAa	0.0050625	0.0082181	0.0031556	
cons	1.585532	2.406244	0.820712	0.000

Table 9: Differences of Standard Error after applying Robust

2. **GLS**

The second solution corrects for heteroskedasticity and autocorrelation by accepting it through applying the GLS model, with a 3-stage model: heteroskedastic, correlated errors across panels and first-order autocorrelation within panels. More details will be presented in Appendix 05.

V. Estimation coefficience

Variables	Coefficient	Probability	Significance
r	-0.0063156	0.355	Insignificant at 1%, 5% level
Infla	-0.0071488	0.219	Insignificant at 1%, 5% level
lngdp	1.050889	0.001	Significant at 1%, 5% level
lnm1	0.1166701	0.137	Insignificant at 1%, 5% level
lnrev	0.2821562	0.005	Significant at 1%, 5% level
lncapex	0.0739457	0.043	Insignificant at 1%, significant at 5% level
eps	0.00000881	0.716	Insignificant at 1%, 5% level
divbm	0.0697323	0.448	Insignificant at 1%, 5% level
ma	-0.0699742	0.080	Insignificant at 1%, 5% level
ROAE	0.0030546	0.485	Insignificant at 1%, 5% level
ROAA	-0.0007701	0.925	Insignificant at 1%, 5% level
_cons	-10.55177	0.000	Significant at 1%, 5% level

Table 10: Beta Coeffience, probability and significance of fixed model

Overall, only 3 out of 11 variables are statistically significant. Specifically, a growth of 1% in GDP will increase SP by 1,051%. In addition, NR and CAPEX are also positively correlated to the SP by 0.28% and 0.07% respectively when they increase by 1% in value. However, CAPEX is only statistically significant at 5% levels.

J. Conclusions & Recommendations

The study uses REM to estimate the impact of several macroeconomic factors, financial indicators and dummy variables related to business activities on the SP and see whether this method is valuable or effective when analyzing panel data. The increasing significance of financial markets has spurred academic interest in the effects of stock market progress on EG. Our study examines the dynamics between stock market progress and economic growth by utilizing the variables M&A and Dividend paid behaviors (dummy variables) in the Vietnam's stock market. Under the development of considering EPS and Dividend payment variables (Fitriana, 2022), the research observed for more years, result in a more objective finding. Following the application of the Hausman test, Wald test, Breusch and Pagan LM test, Autocorrelation Tests, and the implementation of error correction procedures, the outcome involved the utilization of the final correlation result. To be more specific, the correlation between GDP and SP of the group is similar in the positive relationship results of Singh et al. (2011), while there is a no relationship between INF to SP, which is different from the previously researched correlation of of Mulyanti et al. (2019). Moreover, the positive relations between CAPEX and SP are statistically significant (Jiang et al., 2006), which is similar to NR (Srinivasan, 2012). However, the results of the implementation detect a correlation between RIR and M1 with SP as in the study of Stracca (2003). In addition, this research's performance difference in correlation between ROAA and ROAE on SP when they are both determined to be significant (Bayhaqiyi, 2022). Although this paper has different results from previous studies, the results of the correlation of macro variables, financial information and dummy variables are related to SP.

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L. Appendices

I. Appendix 01: The correlation matrix between independent variables (Source: Author's calculation by Stata)

	r	infla	lngdp	lnm1	lnrev	lncapex	eps	bm	ma	ROAE	ROAA
r	1.00										
infla	-0.36	1.00									
lngdp	-0.21	-0.58	1.00								
lnm1	-0.11	-0.05	0.30	1.00							
lnrev	-0.07	-0.17	0.28	0.07	1.00						
lncapex	0.02	-0.05	0.04	-0.02	0.51	1.00					
eps	0.07	0.00	-0.04	0.04	0.23	0.14	1.00				
divbm	0.02	0.04	-0.02	-0.03	0.09	-0.16	0.18	1.00			
ma	-0.03	-0.04	-0.02	-0.02	-0.08	0.29	-0.09	-0.10	1.00		
ROAE	0.05	0.05	-0.07	0.03	0.29	0.14	0.92	0.15	0.11	1.00	
ROAA	0.08	0.02	-0.12	-0.04	0.03	0.07	0.86	0.13	-0.05	0.86	1.00

Table 11: The correlation matrix between independent variables

Summary:

Most correlation values are less than 0.5, some are also negatively correlated with each other. However, the correlation between EPS and both ROAA and ROAE is significant [p(roaa,eps) = 0.92); p(roae,eps) = 0.86]. In addition, the correlation between these two profitability indices is also proportionally cross-correlated.

II. Appendix 2: Explanation for expected signs for independent variables

Independent variable name	Explanation
r	In economic recession conditions, RIR increases, making investments more expensive and less profitable and vice versa. In fact, overall investments are reduced due to the government's way of attracting positive cash flow by imposing high RIR on loans. Since then, investors have become less interested in buying financial assets in general and stocks in particular. Finally, as a basic principle of an economy, the market SP will be positively correlated with its demand.
gdp	GDP growth is a sign of EG. Investments are counted in an economy's GDP. At that time, investors will be more interested in purchasing financial assets. Demand on share will increase and will be proportional to its SP.
infla	Two different opinions are that INF can be both positively and negatively proportional to SP, so the team has no expectations for a correlation between INF and SP.
m1	The increase in total M1 can come from many factors, including reduced RIR to increase EG and promote investment. The amount of M1 will be proportional to financial investments in general and stocks in particular.

	The increase in SP is due to increased investment demand.
rev	The growth in total NR earned before capital costs and OPEX cannot determine whether the company is profitable or not because it is also based on many other cost factors. However, growth in NR shows the company's profitability and the profits that can be generated from it. Furthermore, although the DPR is based on each company's policy, the generation of NI will affect this ratio. In the other words, if this ratio is high, it will attract more investors to invest in the company's stock, promoting the increase of its own demand and leading to the growth of enterprise value.
capex	The investment in TFA may be profitable due to NR growth thanks to increased production. As mentioned, the correlation between NR and SP is positive.
roaa, raoe and eps	Specifically, because the growth of ROAA is based on the profitability of a company's assets, ROAA is included to test its correlation with SP. It is similar to ROAA; ROAE is the profitability of assets that do not come from liabilities. In addition, because EPS is calculated based on NI and share outstanding, an increase in NI leads to an

	increase in the value of the stock. EPS is included to confirm how it correlates to the SP.

Table 12: Explanation for expected signs for independent variables

III. Appendix 03: Hausman Test Details

	(b - B) Difference	sqrt(diag(V_b-V_b)) SE
r	.0006284	.
lnfla	-.0001997	.
lngdp	-.0844825	0.0398429
lnm1	-.001358	.
lnrev	.0507188	0.0297063
lncapex	-.0069547	0.0062348
eps	-3.77e-06	3.47e-06
divdm	-.0028516	.0049269
ma	.0014348	.0037239
roae	.0015385	.0004743
roaa	-.0014008	.0012593

b = Consistent under H0 and Ha;

B = Inconsistent under Ha, efficient under H0;

Test of H0: Difference in coefficient not systematic

Chi2 (**10**) = (b-B)[(V_b - V_B)^(-1)](b-B) = **15.12**

Prob > Chi2 = **0.1276**

(V_b - V_B is not positive definite)

Table 13: Detail Hausman Test

IV. Appendix 04: Test for random effect model and serial correlation

Breusch and Pagan Lagrangian multiplier test for random effects

lnp[ent,t] = Xb + u[ent] + e[ent,t]

Estimated results	Var	SD = sqrt (Var)
lnp	0.1631958	0.403975
e	0.0261597	0.1617396
u	0.0929644	0.3049006

Test: Var(u) = 0

Chibar2(01) = 439.72

Prob > chibar2 = 0.0000

Table 14: The details of Breusch and Pagen Lagrangian Test

Random Effects, Two Sided:		Result
ALM (var(u) = 0) = 299.71	Pr > chi2(1) = 0.0000	Use of Random Effects Model
Random Effects, One Sided:		**Result**
ALM (var (u) = 0) = 17.31	Pr > N (0,1) = 0.0000	Use of Random Effects Model

Serial Correlation		Result
ALM (lambda=0) = 15.08	Pr > chi2(1) = 0.0001	There is serial correlation in the residuals of regression model
Joint Test:		**Result**
LM (var(u) = 0, lambda=0) = 454.80	Pr > chi2(2) = 0.0000	Use of REM but have serial correlation in model

Table 15: Details of test for random effect model and serial correlation

V. Appendix 05: The 3-stages GLS Model to correcting the Violation (Solution 2)

Independent Variables	Coefficient	Probability	Significance
r	-.0132107	0.107	Insignificant at 1%, 5% level
Infla	-.0213337	0.019	Significant at 5% level
lngdp	.9530039	0.000	Significant at 1%, 5% level
lnm1	.1007524	0.124	Insignificant at 1%, 5% level
lnrev	.1291457	0.010	Significant at 5% level
lncapex	.0789396	0.008	Significant at 1%, 5% level
eps	.000004	0.804	Insignificant at 1%, 5% level
divbm	.0952781	0.024	Significant at 5% level

ma	-.0190339	0.508	Insignificant at 1%, 5% level
ROAE	.0004904	0.891	Insignificant at 1%, 5% level
ROAA	.0042299	0.333	Insignificant at 1%, 5% level

Table 16: 3-stage GLS Regression Results.

The GDP and Capex variables have a positive significant impact on the stock market. When one percent changes in GDP, the stock market price increases by 0.9530039% while changing one percent in Capex, SPs 0.0789396%. In contrast, various variables such as interest rate, money supply m1, eps, M&A behavior, ROAE and ROAA are not statistically significant at the 1% and 5% levels. In addition, the remaining variables are statistically significant at the 5% level.